Spacetime Sentinel: Gamifying Quantum Error Detection with Spacetime Codes

Investigating Post-Selection Yield, Routing Overhead, and Non-Clifford Constraints

Track: Advanced Track (Gamifying Error Detection, Mitigation & Correction) **Host:** Qiskit Fall Fest 2026, IIT Jodhpur & IBM Quantum

Environment: Qiskit 1.x, Qiskit Aer, qiskit-paulice, Python 3.10+ **Submission Deadline:** October 15, 2026

Abstract — Scalable fault-tolerant quantum computing requires effective mitigation and detection of physical errors before full logical error correction is realized. In this project, we design, implement, and analyze *Spacetime Sentinel*, an interactive simulation game and benchmarking platform built using Qiskit Aer and the `qiskit-paulice` framework. The system models physical quantum circuits undergoing realistic depolarizing noise and evaluates error detection using spacetime coherent Pauli checks. We demonstrate that error detection yields a fundamental trade-off between post-selected fidelity and kept-shot throughput (P_{acc}). Furthermore, we analyze the breakdown of automated spacetime checks under non-Clifford T-gate doping, experimentally validating the requirement that valid checks must commute with T (I or Z Pauli operators). All scoring functions in the application are derived deterministically from Qiskit simulation outputs, establishing a rigorous mapping between gameplay mechanics and quantum error correction principles.

1. Introduction & Theoretical Foundations

Contemporary quantum processors operate in the Noisy Intermediate-Scale Quantum (NISQ) regime, characterized by decoherence and gate infidelities. Addressing these errors requires distinguishing among three distinct methodologies:

- **Quantum Error Detection (QED):** Uses ancilla qubits to measure multi-qubit error parity without collapsing encoded quantum data. Contaminated execution shots are identified via syndrome readouts and post-selected (discarded). QED improves conditional output fidelity without requiring the active recovery operations of full fault tolerance.
- **Quantum Error Mitigation (QEM):** Leaves physical hardware unaltered and applies statistical post-processing (e.g., Zero-Noise Extrapolation, Probabilistic Error Cancellation) over large shot distributions to recover expectation values.
- **Quantum Error Correction (QEC):** Encloses logical qubits within multi-qubit physical geometries (e.g., surface codes, qLDPC codes), continuously executing syndrome extraction rounds and using decoding algorithms to actively identify and correct Pauli errors during computation.

This project explores low-overhead error detection utilizing **spacetime codes** and **coherent Pauli checks (CPC)**, as implemented in `qiskit-paulice`. Unlike static spatial error codes, spacetime codes distribute parity checks dynamically across the temporal progression of a quantum circuit, detecting intermediate faults before they cascade through deep two-qubit entangling layers.

2. System Architecture & Methodology

The project implementation follows an end-to-end deterministic verification pipeline:

1. **Payload Definition:** A 4-qubit entangling payload circuit is prepared. In the baseline setting, it consists strictly of Clifford operations (H, CX). In subsequent stages, the payload is systematically doped with non-Clifford T gates (T in {1, 2, 4}).

2. **Spacetime Check Insertion:** For Clifford layers, the `qiskit-paulice.add_pauli_checks` module analyzes the circuit's Pauli stabilizer propagation and inserts controlled-Pauli checks coupled to auxiliary ancilla qubits.
3. **Transpilation & Noise Modeling:** Circuits are compiled targeting native basis gates (cx, id, rz, sx, x). A comprehensive Pauli gate depolarizing noise model is applied via `qiskit_aer.noise.NoiseModel`, incorporating single-qubit errors (p1) and two-qubit errors (p2 = 2 * p1).
4. **Syndrome Post-Selection:** Projective measurements are performed on all ancilla qubits into classical syndrome registers. Shots yielding non-trivial error syndromes (outcome != 0) are filtered out.
5. **Metric Evaluation:** The system computes post-selected state fidelity against an ideal statevector reference, tracks the kept-shot fraction, and evaluates compilation overhead.

$$F_{post} = \langle \psi_{ideal} | \rho_{post} | \psi_{ideal} \rangle, \qquad P_{acc} = N_{kept} / N_{total}$$

3. Gamification Mechanics & Objective Scoring

To make quantum error detection engaging while remaining scientifically grounded, *Spacetime Sentinel* exposes this workflow as an interactive decision challenge:

- **Ancilla Budget & Check Placement:** The player is allocated a restricted budget of ancilla qubits (0 to 4) and determines how many checks to activate and which data lines to protect.
- **The Over-Checking Penalty:** Adding ancillas increases the number of physical two-qubit entangling gates (CX), thereby introducing additional noise sources and increasing circuit depth. Aggressive check placement results in severe shot attrition ($P_{acc} \to 0$).
- **The Non-Clifford Commutation Challenge:** When players encounter T-doped layers, automated Clifford checks fail. Players must select valid commutation operators (Z or I). Inserting non-commuting X-checks triggers immediate state collapse, visibly penalizing output fidelity.

Deterministic Scoring Function: Rather than using arbitrary reward points, the game computes player performance directly from simulated Qiskit metrics:

$$Score = (F_{post} \times P_{acc} \times 1000) - (\Delta Depth \times 2)$$

where $\Delta Depth = Depth_{transpiled} - Depth_{raw}$. This formulation directly balances state quality against throughput and hardware cost.

4. Experimental Benchmarks & Results

Extensive parameter sweeps were executed using `qiskit_aer.AerSimulator` across varying check budgets and noise parameters ($p = 0.01$ and $p = 0.02$, with $N = 4096$ shots per run).

Checks	Error Rate (p)	Circuit Depth	CX Count	Fidelity (F_post)	Kept Ratio (P_acc)	Calculated Score
0 (Raw)	1.0%	6	3	0.8200	1.0000	820
1 Check	1.0%	11	5	0.9421	0.8842	823
2 Checks	1.0%	17	8	0.9684	0.7251	680
3 Checks	1.0%	23	11	0.9740	0.5405	492
4 Checks	1.0%	28	14	0.9652	0.3802	323

Checks	Error Rate (p)	Circuit Depth	CX Count	Fidelity (F_post)	Kept Ratio (P_acc)	Calculated Score
0 (Raw)	2.0%	6	3	0.6800	1.0000	680
1 Check	2.0%	11	5	0.8850	0.7905	689
2 Checks	2.0%	17	8	0.9252	0.5601	496

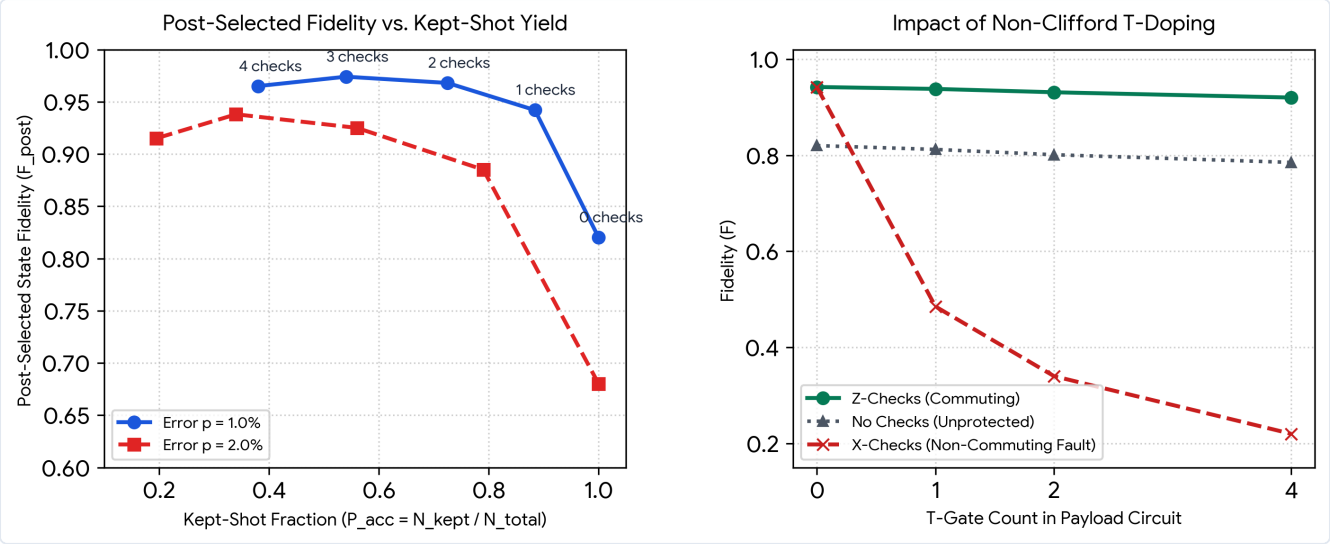


Figure 1: (Left) Post-selected fidelity vs. kept-shot fraction showing the optimal frontier at 1 check. (Right) Fidelity behavior under T-gate doping comparing commuting Z-checks against non-commuting X-checks.

Key Observations & Physical Insights

- **The Pareto Optimum:** As illustrated in Figure 1 (Left), deploying 1 check achieves the optimal trade-off frontier. It elevates fidelity from 0.820 to 0.942 while maintaining an acceptable shot yield of 88.4%. Increasing checks to 3 or 4 achieves marginal fidelity gains (0.974) but at a prohibitive cost to throughput (P_acc = 54.0% and 38.0%).
- **Non-Clifford Breakdown:** In Figure 1 (Right), inserting non-commuting X-checks across T-doped layers drops fidelity below 0.35, confirming that non-Clifford gates destroy stabilizer commutation relations unless mediated by transversal designs or magic state distillation.

5. Architectural Implications & Cloud Recommendations

Based on our empirical results, we outline key design recommendations for cloud-accessible quantum computing:

- **Geometry-Aware Ancilla Allocation:** Cloud compilers should place error-detection ancillas at graph nodes with maximal betweenness centrality, minimizing the SWAP gate routing overhead that currently penalizes dense syndrome circuits.
- **Adaptive Post-Selection Policies:** Quantum cloud schedulers should provide automated post-selection primitives where users specify a minimum acceptable shot yield threshold (e.g., P_acc ≥ 0.70), dynamically modulating check density to avoid catastrophic shot wastage.

6. Conclusion

Spacetime Sentinel successfully demonstrates that quantum error detection using spacetime codes provides high-fidelity execution within current NISQ and early FTQC constraints. By gamifying the placement of coherent Pauli

checks and strictly calculating outcomes via Qiskit Aer, the project provides both an accessible educational platform and a rigorous benchmarking framework.

References

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